

AI JOB DESCRIPTION GENERATOR – BACKEND ARCHITECTURE MAP

(100% User Story Coverage – JD Engine as System of Record)

Note

This is a high-level backend system map for the JD Engine.

JD Engine is a **governed domain service**, not a content generator.

Final scope and rollout depend on phase prioritization.

1. CORE SERVICES

A. API Gateway

- Request routing
 - Authentication & Authorization
 - Rate limiting
 - Request logging
 - CORS handling
 - Payload validation
 - Tenant isolation middleware
 - JD access control enforcement
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B. Authentication & Authorization Service

- User login / logout
 - JWT token issuance
 - Role-based access control (HR, Manager, Approver, Auditor)
 - JD approval permission checks
 - Session & token lifecycle
 - Audit identity propagation
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2. JD DOMAIN SERVICES (CORE BUSINESS LOGIC)

A. Job Analysis Service (Input Layer)

Endpoints

POST /api/jd/job-analysis

GET /api/jd/job-analysis/{id}

PUT /api/jd/job-analysis/{id}

GET /api/jd/job-analysis/{id}/status

Business Logic

- Enforce structured questionnaire only (no free text JD creation)
 - Mandatory fields validation (tasks, skills, reporting line, tools)
 - Draft / Submitted states
 - Lock inputs once AI generation starts
 - Version raw inputs for traceability
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B. JD Generation Service (AI Structuring Layer)

Endpoints

POST /api/jd/generate

GET /api/jd/{jdId}/draft

Business Logic

- Convert job analysis inputs → structured JD model
 - Generate **machine-readable JD**, not prose
 - Normalize skills, competencies, role level
 - Output sections:
 - Job purpose
 - Responsibilities
 - Competencies
 - Proficiency levels
 - KPIs
 - No approval authority here (AI suggests only)
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C. JD Benchmarking Service

Endpoints

POST /api/jd/{jdId}/benchmark

GET /api/jd/{jdId}/benchmark/report

Business Logic

- Benchmark JD against external role libraries (O*NET / ESCO / Industry DB)
 - Detect:
 - Role inflation
 - Missing core competencies
 - Unrealistic experience levels
 - Generate deviation flags
 - Store benchmark dataset version used
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D. JD Validation & Rules Engine

Endpoints

POST /api/jd/{jdId}/validate

GET /api/jd/{jdId}/validation-results

Business Logic

- Deterministic rules override AI output
 - Validate:
 - Role level vs experience
 - Mandatory competencies
 - Reporting consistency
 - Rule priority & conflict resolution
 - Validation must pass before approval
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E. JD Approval & Governance Service

Endpoints

POST /api/jd/{jdId}/submit-approval

POST /api/jd/{jdId}/approve

POST /api/jd/{jdId}/reject

GET /api/jd/{jdId}/approval-history

Business Logic

- HR Manager approval required
- Approval freezes JD content

- Rejection requires reason
 - Approved JD becomes **immutable**
 - Approval events fully audited
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F. JD Versioning & Change Management Service

Endpoints

POST /api/jd/{jdId}/version

GET /api/jd/{jdId}/versions

GET /api/jd/version/{versionId}

Business Logic

- Every change = new version
 - No overwrite allowed
 - Preserve historical references
 - Vacancy & hiring decisions reference JD version explicitly
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G. JD Distribution / Exposure Service

Endpoints

GET /api/jd/{jdId}/export

GET /api/jd/{jdId}/model

GET /api/jd/{jdId}/consumer-access

Business Logic

- Expose JD as machine-readable JSON model
 - Consumer engines:
 - AI Recruiter
 - Performance
 - TNA
 - Mobility / Promotion
 - Enforce read-only access
 - Version-specific consumption
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3. SUPPORTING SERVICES

A. AI Proxy Service

- JD structuring
 - Competency normalization
 - Benchmark comparison
 - Explainability payloads
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B. Rules Configuration Service

- Business rule definitions
 - Competency frameworks
 - Role level mappings
 - Configurable per tenant
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C. Audit & Compliance Service

- JD lifecycle logging
 - Approval traceability
 - Version lineage
 - Regulatory defensibility
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D. Notification Service

- JD approval requests
 - Rejection notifications
 - Change impact alerts
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4. DATABASE ARCHITECTURE

Core Tables

A. Job_Analysis_Inputs

- id (PK)
- created_by
- role_title
- department
- tasks (JSON)

- tools (JSON)
 - reporting_line
 - working_conditions
 - status (Draft / Submitted)
 - created_at
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B. Job_Descriptions

- id (PK)
 - current_version_id (FK)
 - status (Draft / Approved / Archived)
 - created_by
 - approved_by
 - approved_at
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C. Job_Description_Versions

- id (PK)
 - jd_id (FK)
 - version_number
 - structured_model (JSON)
 - benchmark_summary (JSON)
 - validation_results (JSON)
 - created_at
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D. JD_Competencies

- id (PK)
 - jd_version_id (FK)
 - competency_name
 - proficiency_level
 - weight
 - is_mandatory
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E. JD_Benchmark_Results

- id (PK)
 - jd_version_id (FK)
 - benchmark_source
 - deviations (JSON)
 - risk_flags
 - created_at
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F. JD_Approval_Logs

- id (PK)
 - jd_version_id (FK)
 - action (Submitted / Approved / Rejected)
 - actor_id
 - reason
 - timestamp
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G. JD_Consumer_References

- id (PK)
 - jd_version_id (FK)
 - consumer_service
 - reference_type (Vacancy / KPI / Promotion)
 - referenced_at
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5. USER STORY COVERAGE

- ☒ Capture Job Analysis Inputs
- ☒ AI Structured JD Generation
- ☒ Industry Benchmarking
- ☒ Deterministic Validation Rules
- ☒ HR Approval & Governance
- ☒ Immutable Versioning
- ☒ Machine-Readable Exposure